

Experiment 3 | FPGA-Accelerated ECG Signal Filtering and Peak Analysis

Event: V-SPACE FPGA Build Challenge 2026 Platform: PYNQ-Z2 / Zynq XC7Z020

Tools: Vitis HLS and Vivado 2025.1.1, PYNQ Jupyter/Python Report: October 2026

Team / Registration numbers: [team to complete prior to submission]

Evidence scope: This document reports already-saved code, implementation reports and a previously executed physical-board Jupyter demonstration. It does not claim a fresh live board run during report preparation.

1. Objective

Design a custom 128-tap, 16-bit Q4.12 FIR filter accelerated in the FPGA programmable logic; connect it through AXI4-Stream and AXI DMA to the Zynq processor; and validate filtered sample correctness using a synthetic sine test and an excerpt of recorded MIT-BIH ECG. On the ARM processor, demonstrate basic candidate peak detection and heart-rate estimation in Python. The goal is hardware/software co-design and measured functional correctness, not clinical-grade arrhythmia classification.

2. Block Diagram and Architecture

ARCHITECTURE RECONSTRUCTED FROM VIVADO SYSTEM.BD

Verified instance and AXI4-Stream names; not a screenshot of Vivado.

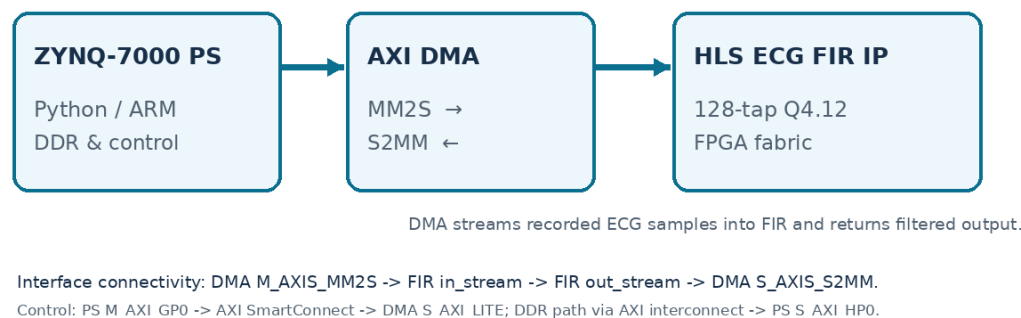


Figure 1. Architecture reconstructed from the recovered Vivado system.bd; this is not a Vivado GUI screenshot.

The recovered Vivado block design contains processing_system7_0, axi_dma_0, ecg_bandpass_filter_0, AXI SmartConnect, AXI memory interconnect and reset/constant support. Verified streams connect AXI DMA M_AXIS_MM2S to the HLS filter in_stream, and the filter out_stream to AXI DMA S_AXIS_S2MM. The Zynq PS controls the DMA over an AXI-Lite path and supplies memory through an AXI high-performance port.

3. RTL Design and Module Hierarchy

The original source is ecg_bandpass_filter.cpp and its header. The original testbench is ecg_bandpass_filter_tb.cpp. Vitis HLS converts this design to synthesizable Verilog; the archive includes the packaged IP component.xml, ecg_bandpass_filter.v, and original HLS project configuration. The Vivado project includes system.bd and system_wrapper.v. Target device: xc7z020clg400-1; HLS synthesis clock target: 10 ns.

Module / file	Verified responsibility
processing_system7_0	ARM software, DDR memory, DMA control
axi_dma_0	Input/output DMA transactions to the FIR stream
ecg_bandpass_filter_0	Synthesized 128-tap FIR HLS IP in programmable logic
system_wrapper.v + system.bd	Block-design integration and FPGA hardware wrapper
ecg_bandpass_filter_tb.cpp	Original HLS C testbench (not a Verilog RTL testbench)

The recovered Vivado project has generated IP-level/out-of-context XDC constraints. A user-authored top-level XDC was not identified in the supplied archives; that fact is recorded in the accompanying constraints provenance note.

4. Simulation Results and Verification

The recovered Vitis HLS C-simulation log reports a passing simulation. The source archive also contains the HLS C-synthesis report. A separate executed Verilog RTL testbench and associated waveform were not found; the original C testbench is provided faithfully and must not be described as RTL simulation.

Test / measurement	Observed saved result
Sine input check	2,048/2,048 samples matched fixed-point reference; max error 0 Q4.12 LSB
5 Hz signal gain	0.9940
50 Hz interference rejection	53.74 dB (as measured in the synthetic sine demonstration)
Recorded ECG replay	12,288/12,288 match; max error 0 LSB; 6 DMA batches
Peak/heart-rate postprocessing	42 peaks; RR 0.812 s; estimated heart rate 73.9 BPM

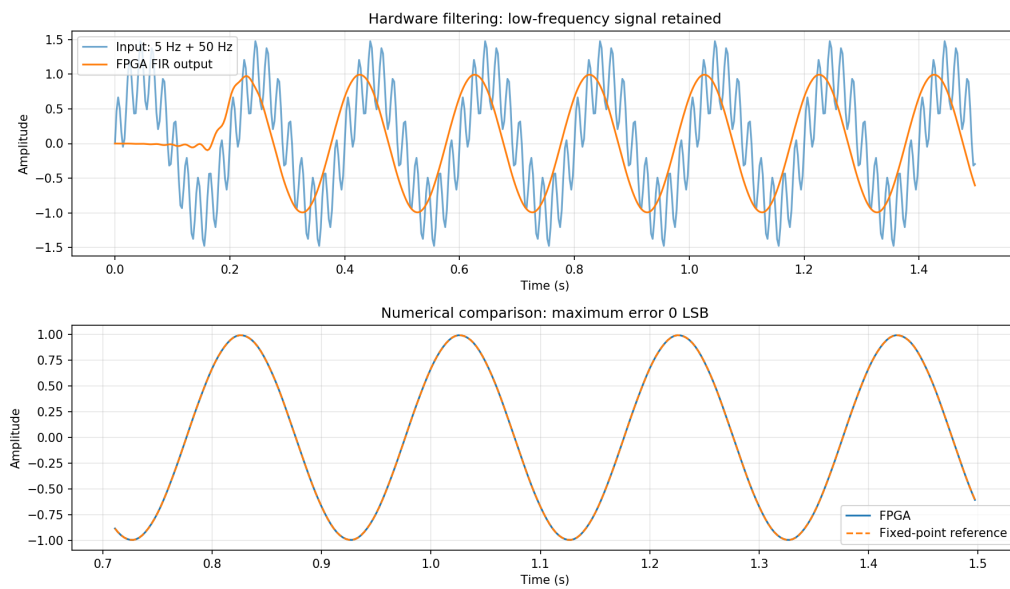


Figure 2. Recovered sine verification plot from the real board notebook outputs.

5. Hardware Implementation and Results

The FPGA overlay loaded successfully on PYNQ-Z2, with IP `axi_dma_0` detected. The recorded ECG sample rate was 360 Hz, with 12,288 samples corresponding to 34.13 seconds and transferred in six batches of 2,048 samples. The project used MIT-BIH record 100, channel MLII, and added a documented 50 Hz sine interference to illustrate filtering.

The original Jupyter output reports Python-plus-DMA wall time of 7.55 ms for the measured run. This is not FPGA-only latency and does not establish sensor acquisition or end-to-end real-time performance. Peaks were found using Python height threshold and a 300 ms minimum separation; this is not an FPGA QRS detector or a clinically validated diagnosis method.

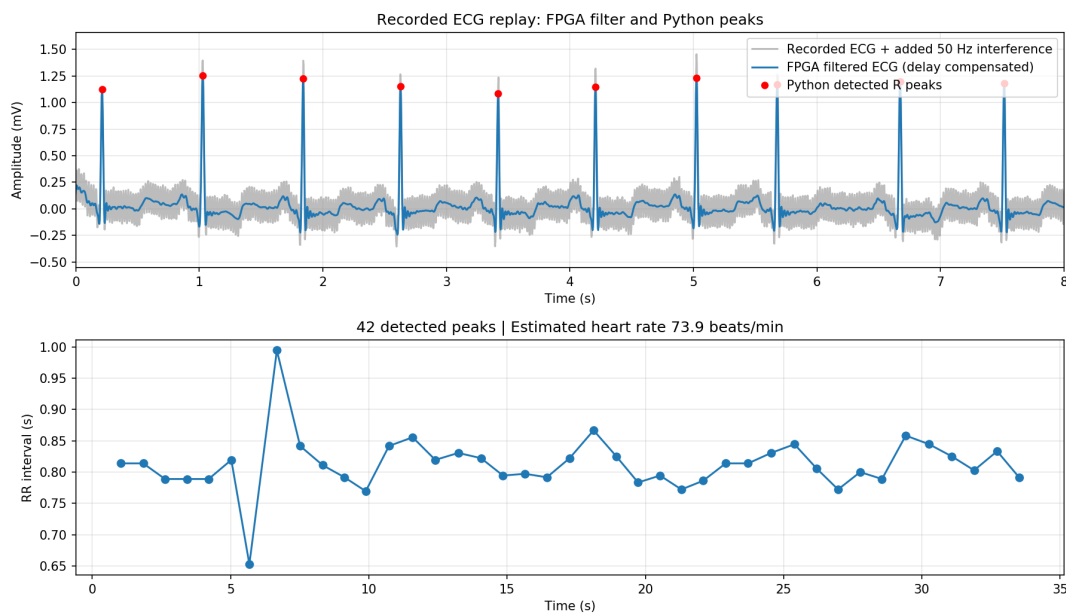


Figure 3. Recovered recorded-ECG trace and red peak annotations computed in Python on ARM.

Vivado placed resource summary from the supplied implementation report: 7,232 slice LUTs, 13,844 slice registers, 2 block RAM tiles, and 40 DSPs. These are whole-project placed counts, not FIR-only utilization.

6. Applications

Potential application areas include educational FPGA-based biomedical signal processing, hardware-accelerated ECG preprocessing for wearable or embedded systems, and mixed PS/PL data-path demonstrations. Clinical use would require sensor interfacing, robust detector validation, safety assessment, and compliance evaluation beyond this project.

7. Conclusion and Future Scope

The recovered HLS implementation, Vivado design, bitstream, HWH metadata and board evidence are consistent. The preserved demonstration verified exact fixed-point agreement for synthetic and recorded ECG replay tests and demonstrated peak/heart-rate processing on ARM. Future scope includes live ADC/sensor acquisition, a hardware QRS stage, controlled performance benchmarking, independently executed RTL simulation evidence, and validated multi-record testing.

Source Provenance and Submission Caveats

Original digital archives: `ecg_bandpass_fir.zip` (Vitis HLS), `project_1.zip` (Vivado); `Project3_ECG_Board_Evidence.zip` (12 original saved board files). Data source: MIT-BIH Arrhythmia Database v1.0.0, record 100 MLII; <https://physionet.org/content/mitdb/1.0.0/> ; ODC Attribution License v1.0 as recorded in the supplied `DATA_SOURCE.md`.

Items not in current evidence: narrated final video URL, a physical board photograph, an actual Vivado RTL schematic screenshot, and a separately verified Verilog RTL testbench/waveform. The reconstructed architecture is explicitly labeled as derived, not an original screenshot. Team metadata must be completed before publication.

Submission-guideline reference: user-provided V-SPACE FPGA Build Challenge 2026 Submission Guidelines, six-page PDF. This report follows its objective, architecture, RTL, simulation, hardware, applications and conclusion headings.